

Valuing a Gaze Decision under Limited Visual Access: Active Sampling and Human Choice

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Abstract

How should an observer value a possible gaze decision when detailed visual access is limited? We study this question through three connected analyses: a full-image behavioral reference, autonomous sampling under partial observation, and conditional models of recorded human choices. The current implementation uses static images and a fixed observation budget. A shared observer predicts full-image visual features from a blurred view with retained clear patches. Local uncertainty, expected local prediction gain, and expected global prediction gain provide three candidate values. In closed-loop experiments on 371 development images, local gain gives the lowest mean terminal feature error; a common movement cap makes terminal differences from uncertainty inconclusive. Separately, four conditional choice models are fitted along human trajectories on training images. A frozen evaluation on 603 new images and 17,837 targets finds higher human next-choice log probability for local gain than uncertainty, by 0.05637 nats, and by 0.04322 nats in a fixed disagreement subset. Both adjusted intervals exclude zero. Full-image DeepGaze III gives the highest human-choice score among the evaluated models, but was not tested as a closed-loop sampling policy. Thus, expected prediction gain provides a useful operational measure of gaze value in this setting, while uncertainty is a competing state-based signal. The results concern predictive utility, not Shannon information or a demonstrated human mechanism. Gain-estimation bias, simplified memory, and the distinction between model and human visual access limit the conclusions.

1 Introduction

The broader project asks how to value a gaze decision in active sensing under limited resources. Here, the resource restriction is concrete: only small regions receive clear visual evidence, and the observer has a fixed number of observations. We do not model a general computational, energy, or duration budget. The question is not only where gaze is likely to land, but what obtaining evidence at a location would accomplish for a particular observer, given its current information.

To make this question measurable, four ingredients must be specified: what the observer can see, what it tries to estimate, how improvement is measured, and which observations are allowed. We define sensory information as evidence available to update a prediction, and *value* as the expected benefit of acquiring that evidence for a stated objective. In this first implementation, the objective is prediction of full-scene visual features and the benefit is a reduction in squared error. This is an operational working hypothesis for free viewing, not an established description of the participant’s goal.

Looking changes the evidence available to a visual observer. A useful next fixation may reveal an uncertain object, confirm a local detail, or provide information about other parts of a scene. These possibilities raise a basic question: should an observer look where prediction is currently most uncertain, or where a new observation is expected to improve prediction most?

The two rules can disagree. A location can be uncertain because its contents remain difficult to measure even after looking. Another location can have moderate uncertainty but be easy to resolve. A third can reveal a relationship that improves predictions elsewhere. Uncertainty describes the current prediction; gain describes the expected effect of an action. Their relationship therefore depends on both the observation process and the dependencies in the scene.

Active-sensing experiments have shown that human fixations can use previously acquired evidence and stimulus statistics in visual categorization (Yang et al., 2016). Free viewing poses a different challenge: the observer has no explicit external task, and the researcher does not know which internal goals drive each fixation. We use a restricted, testable working hypothesis: if looking serves to improve a representation of an image, how well do uncertainty and expected prediction gain explain the next location selected by a person?

We compare three signals inside one fixed observer: local predictive uncertainty U , expected local error reduction G_{local} , and expected whole-image error reduction G_{global} . The local-gain comparison is important. It tests how much local prediction error a new observation can remove without also attributing value to other regions. Global gain adds this second source of value.

The project has three connected branches (Figure 1). **A: Behavioral reference** asks how well the next human fixation can be predicted using the full image and previous fixations. **B: Functional active sensing** asks which value-based sampling policy helps a restricted observer predict the scene better. **C: Human-choice modeling** asks which candidate value predicts the actual next human fixation after common spatial and history controls. Branch A supplies a reference for C. Branch B supplies the common observer and value signals, but its autonomous histories are not used in place of human histories in C.

A rule that improves reconstruction need not predict human behavior; a useful behavioral signal need not win a reconstruction comparison. We therefore do not use closed-loop victory as a condition for the human analysis. Moreover, a sampling policy and a probabilistic choice model are different objects: one selects the model’s next action, while the other assigns probability to a person’s recorded next action.

The current study is the static-image, population-level implementation of this broader program. Earlier proposals also considered video, observer-specific preferences, duration, motion, and smooth pursuit. Those components have not been implemented here. Likewise, the current behavioral reference reuses pretrained DeepGaze III; we did not fit a new individualized reference model or test whether value adds information beyond DeepGaze as a covariate.

Our contribution is a controlled empirical comparison, not a new general theory of information seeking. We (i) demonstrate exact conditions under which the signals agree or disagree, (ii) construct and diagnose a shared partial-view observer, and (iii) compare frozen choice models on new images, including a disagreement subset defined before final evaluation. The strongest result concerns local gain. The final analysis also preserves less favorable evidence: gain magnitudes remain biased, movement constraints weaken terminal reconstruction differences, and a development-set revisit effect does not clearly replicate.

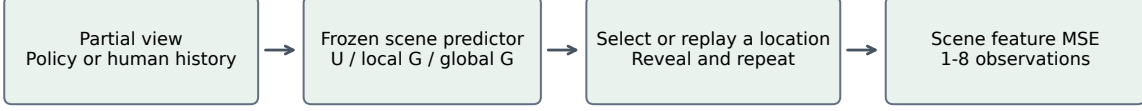
2 Related work

Active sensing connects action selection to the information that an observation can provide. The visual categorization study of Yang et al. (2016) motivates this view, but its task-specific findings do not establish a free-viewing objective. Renninger et al. (2007) distinguish local and global uncertainty-reduction strategies in a controlled shape task. Artificial active-sensing work also couples perception and exploration (Sharafeldin et al., 2024). Dynamic feature selection provides a related acquisition-budget setting, in which additional features are valued for predictive benefit (Gadgil et al., 2024). Our loss-based gain is the expected reduction in feature mean squared

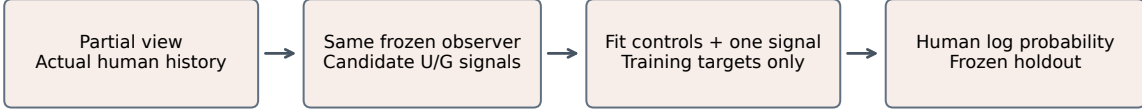
A. Behavioral reference



B. Active sensing under limited visual access



C. Conditional models of human choices



Training order: mean → freeze → independent variance → freeze → gain → freeze → choice coefficients

Resource limits: clear-patch size, observation count, and a shared movement-cap control.

Figure 1: Overall active-sensing framework and the three completed analyses. A uses pretrained full-image DeepGaze III as a behavioral reference. B lets each policy choose its own observations and measures feature error. C follows recorded human histories, fits separate conditional choice models, and measures next-cell log probability. B and C share a frozen observer but use different action histories. The partial view is a modeling assumption; the original participants saw complete images.

error (MSE). It is not automatically equivalent to Shannon information gain, the conditional mutual information studied by [Gadgil et al. \(2024\)](#), or long-horizon utility.

Free-viewing and search datasets support quantitative scanpath modeling ([Chen et al., 2022](#); [COCO-FreeView Dataset](#)). Models such as the Human Attention Transformer use scene content and observation history to predict gaze ([Yang et al., 2024](#)). DeepGaze III provides a learned probabilistic reference for free-viewing scanpaths ([Kümmerer et al., 2022](#)). Here it serves as a full-image descriptive benchmark; it is not a belief state or a pure habitual prior.

We use DINOv2 features as a fixed visual prediction target ([Oquab et al., 2024](#)). Their utility as general visual features does not establish them as a human representation. We also separate mean fitting from variance estimation. This makes the two components easier to diagnose, given known optimization difficulties in heteroscedastic Gaussian likelihood training ([Seitzer et al., 2022](#)). Finally, model recovery checks whether the proposed choice models can be distinguished under their own fitted signal strengths, following the broader need to validate behavioral models before interpreting them ([Wilson and Collins, 2019](#)).

3 Methods

3.1 The three analyses and what is learned

Table 1 separates the inputs, learned components, and outcomes. There are two kinds of prediction: predicting the *scene representation* and predicting the *next human action*. Feature MSE measures the first; conditional next-cell log probability measures the second. They have different units and must not be combined into one ranking.

Training proceeds in stages. First fit the partial-view mean predictor, freeze it, then fit the

Table 1: The completed project, organized by question rather than model name. All final U/G comparisons use the same frozen visual observer.

Branch	Input and history	Learned / fixed components	Outcome and evaluation
A. Behavioral reference	Full displayed image; actual human fixation prefix	Pretrained DeepGaze III and external center bias; no project-specific refit	Human next-cell log probability; matched final targets
B. Active sampling	Partial view; policy history, plus offline human-trajectory replay	Train mean, independent variance, and gain; freeze all; five policies and recorded human paths	Feature MSE over 1-8 observations; 371 development images; 364 matched human comparisons
C. Human choice	Partial view reconstructed from the actual human prefix	Same frozen U/G; fit four separate control/value softmax models on training trajectories	Human next-cell log probability; 603 hold-out images, 17,837 targets

Table 2: Fixed data separation. All participants for an image remain in the same group. Counts include the initial fixation.

Split	Images	Trials	Fixations	Role
Training	3,343	33,340	515,359	Parameter fitting
Development	371	3,698	56,825	Selection and diagnosis
Internal holdout	603	6,010	92,821	Frozen final evaluation

independent variance head. Next generate before/after error changes using that frozen observer and train a gain head. These components are then frozen for both B and C. Finally, C fits the choice-model coefficients on 99,004 supported next targets from 3,343 training images. The human action labels train these choice coefficients; they do not turn the greedy U/G policies into imitation-trained policies. Development data guide upstream checkpoint selection, and the final holdout is used only after the complete specification is locked.

3.2 A. Full-image behavioral reference

The reference is $q_\phi(a_t | I, h_t)$: a probability map from pretrained DeepGaze III, conditioned on the complete image and the recorded human prefix. We reuse its published pretrained parameters rather than train a new reference on COCO-FreeView. Pixel probabilities are integrated into the same content cells used by branch C. This makes the scored targets and outcomes comparable, while preserving the difference in visual access. DeepGaze is neither the partial observer’s internal belief nor a pure motor prior. It is not run as a sampling policy in branch B.

3.3 Data and image-level separation

We analyze publicly available COCO-FreeView records. The original study asked 10 participants to view each image freely for five seconds, without a cued search target or required response (COCO-FreeView Dataset). This work reuses those records and does not collect new participant data. The downloaded train and validation files contain 4,317 distinct images, 43,048 trials, and 665,005 fixations. These are the audited local counts, not the size of the complete dataset including its test portion.

The official training set supplies our training and development groups; the official validation set

supplies the internal holdout (Table 2). A fixed seeded image ordering defines the split. The holdout is not the official test set. Participants occur across image groups, so the evaluation concerns new images for these participants, not generalization to a new participant population. The first recorded fixation is an initial condition and is never scored as a free choice. We preserve its measured location rather than replacing it by an exact center. Images already contain the experimental display borders. We retain all border fixations in observation histories and exclude only targets outside the image content from content-cell scoring. Twenty overlays were used to inspect image identity, coordinates, and fixation order.

3.4 B. Restricted observation and scene prediction

Let I be an image and $h_t = (a_0, \dots, a_{t-1})$ its first t recorded or selected fixation locations. The partial view $O(I, h_t)$ has a blurred background and clear circular patches around these locations. Previously revealed pixels remain clear. The patch radius is 8% of the shorter content dimension; blur scale is 5%. The display is resized to 448×280 pixels. Content masks separate image content from existing display borders and any encoder padding.

This is a simple access-and-memory model. The original participants saw full images; we impose partial access only in the computational replay. Blur preserves coarse scene information, so unrevealed regions are not completely unobserved. Perfect retention of clear patches is also an assumption, not a measured property of human memory.

A frozen DINOv2 ViT-S/14 encoder maps the partial RGB image to a 20×32 grid with 384 channels. The target $z = \phi(I)$ is the corresponding full-image feature grid, normalized using training statistics. A mean network predicts $\mu(h_t)$ and an independent variance network predicts $v(h_t)$. Crucially, blurring and revealing occur in RGB space *before* feature extraction. Full-image target features are available only for training and offline error measurement.

3.5 Uncertainty, realized change, and expected gain

For a candidate fixation a , let $R(a)$ denote its circular content footprint. For any region R , let $w_{R,p}$ be the fraction of encoder patch p covered by that region, and let $C = 384$. We define

$$L_R(h; z) = \frac{\sum_p w_{R,p} \sum_{c=1}^C (\mu_{pc}(h) - z_{pc})^2}{C \sum_p w_{R,p}}, \quad U(a, h) = \frac{\sum_p w_{R(a),p} \sum_{c=1}^C v_{pc}(h)}{C \sum_p w_{R(a),p}}. \quad (1)$$

The full-content loss L_Ω uses the valid content mask. The hypothetical new state $h \oplus a$ includes the patch revealed at a . Its realized gains are

$$\Delta_{\text{local}}(a, h; z) = L_{R(a)}(h; z) - L_{R(a)}(h \oplus a; z), \quad (2)$$

$$\Delta_{\text{global}}(a, h; z) = L_\Omega(h; z) - L_\Omega(h \oplus a; z). \quad (3)$$

The local comparison uses the same candidate footprint before and after observation, including any previously visible pixels within it. It is distinct from a diagnostic restricted to newly revealed pixels.

The corresponding action values are approximations to conditional expected gains,

$$G_R(a, h) \approx \mathbb{E}[\Delta_R(a, h; z) \mid O(I, h), h, a]. \quad (4)$$

A separate network learns these expectations from realized labels. At inference it receives only summaries available before looking, never the target or the after-view. A single realized gain may be negative. Our approximate, frozen predictor is not an exact Bayesian posterior mean, so even its expected gain need not be nonnegative. Negative labels and predictions are not clipped.

Predictive variance should also be interpreted carefully. With a fixed imperfect mean, Gaussian likelihood training estimates conditional squared residuals, which may include mean bias. We use U as an operational predictive-uncertainty signal; it is neither a validated subjective uncertainty nor a separation of epistemic and aleatoric uncertainty.

3.6 Sequential fitting and diagnostic checks

We first fit the mean with content-weighted MSE. After a bounded extension with learning-rate decay and early stopping, training stops at epoch 70 and selects epoch 60. We then freeze the mean and all parameters affecting it, and train an independent variance head for 20 epochs using

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left\langle \log(2\pi v) + \frac{(z - \mu)^2}{v} \right\rangle_{\Omega}. \quad (5)$$

This sequence follows development diagnostics of an earlier joint mean/variance model; it is not a claim that joint likelihood training is generally invalid.

After freezing the new observer, we regenerate all gain labels. The gain dataset uses human and random histories at budgets $t \in \{1, 4, 7\}$, with 160,464 training candidates and 35,616 development candidates. A two-output network predicts local and global gain jointly. It trains for 10 epochs; development global-gain MSE selects epoch 2. All later analyses use this same checkpoint, including the local-gain analysis. No old-observer gain model is reused. Appendix A gives architecture and optimization details.

Diagnostics test both whether the mean benefits from new evidence and whether variance and gain have useful scale and ranking. Near-zero change alone would be insufficient: a predictor could ignore a new patch. We therefore compare correct reveals with a control that updates visibility without supplying the correct new visual evidence. Development diagnostics support actual use of the patch, while also exposing remaining calibration errors.

3.7 B. Policies select the model’s own next observations

For signal $s \in \{U, G_{\text{local}}, G_{\text{global}}\}$, the greedy policy is $a_t = \arg \max_{a \in \mathcal{A}(h_t)} s(a, h_t)$, with fixed tie handling. The candidate set $\mathcal{A}$ enforces new-pixel eligibility and, in the constrained experiment, the common movement cap. The selected location changes the next partial view. The predictor is not retrained between policies or steps.

Five policies select candidate cells: U , G_{local} , G_{global} , uniform sampling, and a center preference estimated from training data. The three signal policies greedily select their highest-scoring eligible location. The random baselines each use five seeds. Each image has shared initial conditions, and each policy then accumulates its own observations for eight budgets. Candidates revealing no new pixels are excluded in this autonomous task.

We evaluate all 371 development images, first without a movement limit and then with a shared single-step distance cap. The cap is 0.20785 times the shorter content dimension, determined from training trajectories: the median across images of each image’s median valid consecutive movement. All policies use the same cap. A shared cap does not force equal total movement or equal revealed area. The closed-loop evaluations are development results, not final holdout reconstruction tests.

3.8 B. Offline human paths as scene-prediction inputs

A later development analysis replays the first 1–8 actual human fixations through the same frozen observer. Observation 1 is the experimental starting fixation; observation 8 therefore

adds seven observations. The primary comparison uses the participant already selected for each image by the original closed-loop experiment. Human and policy paths share that participant’s starting point. Seven selected human trials have fewer than eight fixations; we do not pad them. All primary curve points use the same remaining 364 images.

Each human path is evaluated separately. We then average its MSE equally across images. For each random policy, its five seeds are averaged within image first. A supplementary human curve uses all 3,620 complete eight-fixation prefixes across 371 images: average participant MSE within each image, then average images equally. It excludes 78 short trials and has different participant starting points, so it is not a substitute for the primary paired-start comparison. Neither analysis averages fixation coordinates, merges participants’ revealed regions, or averages their predicted features before computing error.

Human coordinates remain continuous, with original revisits and border points. Policies use grid locations and exclude no-new-pixel candidates. We replay the saved unrestricted policy paths and rescore them in the same process and device as the human paths, checking starting MSE and coverage, rather than mixing newly computed human errors with old policy errors. We do not regenerate policy decisions. This controls scoring-environment differences within this comparison; it does not establish exact reproduction of the original GPU execution. The initial attempts failed the strict same-start check against old scores, and are retained as failures. Coverage and cumulative movement are reported, not equalized. All contrasts use 2,000 paired-image bootstrap replicates with descriptive 95% intervals and no multiplicity adjustment. This analysis follows the original holdout evaluation and uses development images only.

3.9 C. Fit probabilities along actual human trajectories

Branch B also includes a later offline replay of human paths, described above. That analysis asks how human-selected evidence helps the model predict the scene. Branch C instead asks how well a model predicts the next human action.

Human replay uses the actual first t recorded fixations to predict target a_t , for $t \in \{1, 4, 7\}$. These budgets lie within observer training support and match gain-label budgets. We retain genuine returns and the full content-cell grid. A previously visited cell is not necessarily a location with zero new pixels. Under exact repeated observations with no new pixels, the perfect-memory model has unchanged input and zero gain; these candidates remain in human scoring, with their uncertainty values retained.

The actual next target is a training label or an evaluation target; it is not used to construct its own before-choice U/G map. We fit a conditional multinomial-logit model (McFadden, 1974) at the candidate-cell level. We do not generate a free-running scanpath and match its shape to the recorded trajectory. We also do not assign human likelihood to the deterministic argmax policy from B. Instead, learned coefficients convert candidate signals and controls into a full probability distribution.

The candidate grid has 16 cells along the longer content dimension, with the other dimension adjusted to content aspect ratio. This is distinct from the encoder grid. The four separately fitted models are **Controls**, **Controls+U**, **Controls+G_{local}**, and **Controls+G_{global}**. They use the same control variables and fit different coefficient vectors. For signal s ,

$$p_s(a \mid h) = \frac{A_a \exp[\beta_s^\top \tilde{\mathbf{c}}(a, h) + \gamma_s \tilde{s}(a, h)]}{\sum_b A_b \exp[\beta_s^\top \tilde{\mathbf{c}}(b, h) + \gamma_s \tilde{s}(b, h)]}, \quad (6)$$

where A_a is the cell’s fraction of content area. The control-only model omits the signal term. The 11 controls describe content position, movement distance, new visible area, revealed fraction, and previous visits (Appendix B). Standardization and fitting use training images only. Coefficients

are unconstrained; a negative signal coefficient would not indicate preference for high values. The four models are fitted separately with fixed L_2 regularization of 0.001.

DeepGaze III reads the full displayed image and the same actual fixation prefix. Its pixel probabilities are summed into the same candidate cells and conditioned on image content. It is scored on exactly the same targets as the choice models, along with its external center-bias reference. Thus, the original full-trajectory DeepGaze average is not compared to our restricted-budget average. Possible overlap between DeepGaze’s pretraining images and COCO remains unresolved.

3.10 Final protocol and uncertainty intervals

Before holdout scoring, we fix model weights, standardization, scoring code, and two primary local-gain versus uncertainty comparisons. The first uses every supported content target. The second uses states where the two signals have different unique top-ranked candidates, their new visible areas differ by at most 0.002 of total content area, and their movement distances differ by at most 0.05 of the shorter content dimension. Subset membership uses only signals and history available before the next choice.

These thresholds came from development exploration and were locked before final scoring; this is not a preregistration before data collection. Matching two preferred candidates approximately on area and distance is also not full causal matching of all candidates or histories.

We average log probabilities across targets within each image and then equally across images. Paired bootstrap intervals resample images, with 2,000 replicates. Each of the two primary comparisons has a two-sided 97.5% interval, giving a Bonferroni-adjusted family coverage of 95%. All other intervals are descriptive 95% intervals, without a family-wide correction. Image resampling does not establish uncertainty over a population of new participants.

Model recovery uses each fitted choice model to generate choices on recorded candidate states, followed by refitting all four models. Thirty replicates per generator yield 480 fits. A highest mean score is reported separately from a winner supported against alternatives by six-pair Bonferroni-corrected intervals. This recovery holds histories fixed; it does not simulate new scanpaths or establish recovery at arbitrary signal strengths.

4 Results

4.1 Three answers at a glance

Which policy helps the observer predict the scene? Local gain has the lowest mean feature MSE after eight observations among the five tested policies. Without a movement cap, both gains improve on U with supported paired differences. Under the common cap, the terminal differences from U are inconclusive, while path-average differences remain negative. These are development-image functional results.

Which fitted value model predicts human choices? On the frozen holdout, Controls+local gain gives the highest human-choice log probability among the three value models, followed by Controls+global gain and Controls+U. The conditional models were actually fitted on training trajectories, then scored along real held-out prefixes.

How does DeepGaze III compare? It gives the highest human-choice log probability among all evaluated models: -3.854297 , compared with -4.138800 for Controls+local gain. This is a full-image behavioral reference comparison. No DeepGaze closed-loop MSE exists in this study because it was not run as a sampling policy. Sections below report the diagnostics and exact comparisons behind these answers.

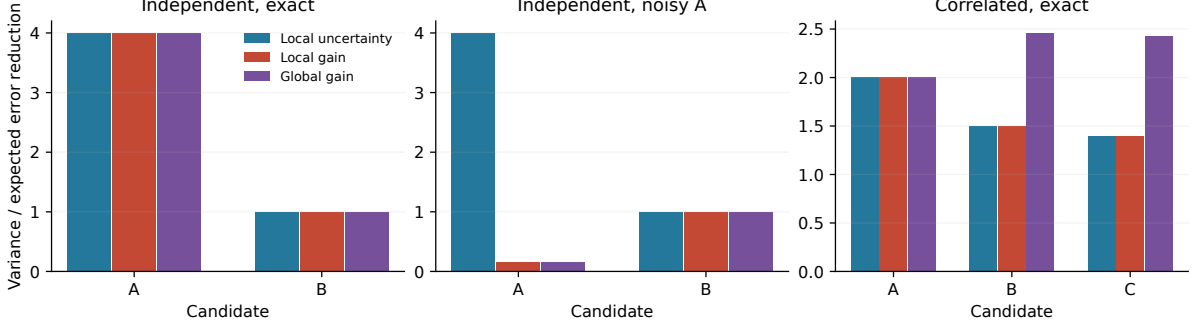


Figure 2: Exact Gaussian examples. Left: independent variables with exact observation. Middle: A is uncertain but difficult to measure. Right: B and C are correlated, making B useful beyond its own location. Global gain is shown as reduction in summed error, not mean error; comparisons of policy rankings are within panels. Bars are analytic values from previously validated examples.

4.2 Exact examples separate uncertainty from observation value

For a Gaussian state $z \sim \mathcal{N}(0, \Sigma)$ and observation $y = z_a + \epsilon$, with independent noise variance τ_a^2 , exact conditional-mean prediction gives

$$U(a) = \Sigma_{aa}, \quad G_{\text{local}}(a) = \frac{\Sigma_{aa}^2}{\Sigma_{aa} + \tau_a^2}, \quad G_{\text{global, sum}}(a) = \frac{\sum_j \Sigma_{ja}^2}{\Sigma_{aa} + \tau_a^2}. \quad (7)$$

The last expression uses summed squared error to expose contributions from other variables. Dividing by the number of variables gives global MSE gain without changing the ranking within a case.

Figure 2 shows three fixed examples. Independent, noiseless variables produce identical rankings. Large observation noise at candidate A reverses the gain ranking despite high uncertainty there. Correlation between B and C lets B improve another variable, so global gain favors B while uncertainty and local gain favor A. In the correlated case, B has 1.5 units of local gain and 0.96 additional units from C. Each case was checked with 200,000 Monte Carlo draws; all 14 simulated gain means matched their analytic values within the fixed five-standard-error tolerance. These examples validate the distinction and implementation, not a claim about human fixations.

4.3 The observer uses new patches, but gain estimation is imperfect

The final mean achieves development MSE 0.73222, compared with 0.99762 for the training-mean feature baseline on the same states. Across 17,808 diagnostic candidates, revealing a patch improves error in newly revealed pixels by 0.02475 on average, with a 95% image interval of [0.02204, 0.02742]. The correct-patch versus visibility-only control also favors correct evidence, with mean new-region improvement 0.02789. This supports effective use of visual input rather than simple insensitivity to a reveal.

All 2,226 checked development mean outputs remain identical during independent variance training. The ratio of candidate predicted variance to observed squared error is 0.9675, indicating about 3.3% aggregate underestimation. Within-state uncertainty/error rank correlation is approximately 0.4295. These checks support a usable operational signal, not perfect conditional calibration or Gaussian coverage at every location.

Local gain has useful ranking and prediction accuracy, but the global-gain MSE advantage over a constant is uncertain (Table 3). Both underestimate gain magnitude. About 38.1% of local realized gains are negative despite a positive average. We therefore interpret subsequent comparisons as tests of fixed, imperfect predictive signals. A trainable choice coefficient can

Table 3: Gain quality on development candidates. MSE differences are relative to a constant predictor; negative is better. Both gain estimates underestimate average realized improvement.

Measure	Local gain	Global gain
Within-state rank correlation	0.2508	0.1092
Mean predicted gain	0.006427	0.001564
Mean realized gain	0.024484	0.003249
MSE minus constant baseline	−0.000769	−0.000006
95% interval for MSE difference	[−0.000898, −0.000630]	[−0.000017, 0.000004]

Table 4: Terminal feature MSE after eight observations on the same 371 development images. Lower is better. These are policy outcomes, not human likelihoods. Local gain has the lowest point estimate; terminal G–U intervals under the common cap include zero.

Sampling policy	No movement cap	Common movement cap
Uncertainty U	0.718323	0.718316
Local gain	0.711176	0.714982
Global gain	0.713069	0.716672
Uniform	0.720838	0.721623
Training center preference	0.716785	0.721130
DeepGaze III	Not evaluated as a sampling policy	

rescale a signal but does not repair its ranking errors or prove that its numerical magnitude is calibrated.

4.4 B. Which sampling policy improves scene prediction?

Without a movement cap, terminal development MSE is 0.71832 for U , 0.71118 for local gain, and 0.71307 for global gain. The paired local-gain minus uncertainty difference is -0.00715 , with a 95% interval of $[-0.01124, -0.00318]$. The global-gain difference is -0.00525 , with interval $[-0.00949, -0.00118]$. Local versus global terminal differences are not clearly separated. Coverage also differs: 9.49% for uncertainty, 10.70% for local gain, and 10.16% for global gain.

The common movement cap weakens the terminal comparison (Figure 3). Local-gain minus uncertainty terminal MSE is -0.00333 with interval $[-0.00724, 0.00072]$; the global counterpart is -0.00164 with interval $[-0.00581, 0.00231]$. Both include zero. Average MSE over budgets 1–8 remains lower for local gain by 0.00358 and global gain by 0.00254, with negative paired intervals. No policy stalls under the cap. Actual distance and coverage remain unequal, so these are policy-level functional effects under a common constraint, not effects with exactly matched movement and area.

4.5 B. Do recorded human paths improve scene prediction?

Yes, under offline replay: human-selected observations reduce the frozen model’s mean MSE from 0.745708 to 0.718516 on the 364 matched images, a reduction of 0.027192 with descriptive 95% paired-image interval $[0.022662, 0.031821]$. Table 5 and Figure 4 compare human paths with the saved unrestricted policies, all rescored in the same environment. Their sample and scoring procedure differ from the original 371-image policy experiment; its reported results remain unchanged.

At observation 8, human-minus-local-G MSE is 0.007091, $[0.002967, 0.011362]$, and human-

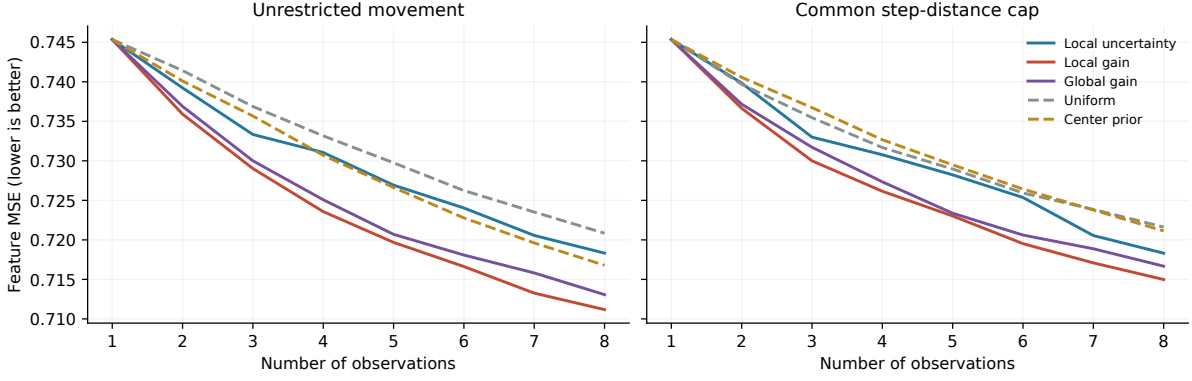


Figure 3: Closed-loop prediction on 371 development images. Curves show image-averaged error; lower is better. Policies start from the same initial condition and then acquire different evidence. The shared distance cap reduces terminal separation. Curves do not display confidence bands; paired terminal and path-average intervals are reported in the text and Appendix C.

Table 5: Scene prediction from human and policy paths on 364 common development images with matched starting points. All start at MSE 0.745708. Reduction is MSE at observation 1 minus MSE at observation 8; coverage is the revealed content fraction. Movement is cumulative distance in units of the content’s shorter side.

Trajectory	MSE at 8	Reduction 1–8	Coverage	Movement
Human	0.718516	0.027192	8.58%	1.717
Uncertainty U	0.718354	0.027354	9.49%	2.338
Local G	0.711424	0.034283	10.70%	2.947
Global G	0.713309	0.032399	10.16%	2.682
Center prior	0.716930	0.028777	10.31%	2.931
Uniform	0.721008	0.024700	10.54%	4.271

minus-global-G is 0.005207, [0.001044, 0.009385]: both gain paths have lower model error under these descriptive intervals. Human differences from U, center, and uniform sampling each have intervals containing zero. Averaged over observations 1–8, human MSE is 0.729578; it is lower than uniform sampling by 0.002859, [−0.005116, −0.000431], but higher than local G by 0.004892, [0.002002, 0.007754], and global G by 0.003638, [0.000821, 0.006526]. Path-average differences from U and center remain inconclusive. These multiple comparisons are exploratory.

The all-participant supplement gives terminal MSE 0.717939, after averaging errors within image and then across images. Human paths in the primary comparison reveal less area and travel less than the tested policies. Participants originally saw ordinary full images, not this simulated blur-and-reveal interface. These findings measure the usefulness of their recorded locations for this frozen model, not human internal prediction error or evidence that people are worse at optimizing their own goals.

4.6 A and C. Which models predict the human next choice?

The fixed final evaluation scores 17,837 targets from all 603 holdout images. Of three possible budgets for each of 6,010 trials, 180 targets are unavailable because a trial is too short and 13 lie outside image content. All four choice models and both references use the same included targets. No parameters are fitted in this evaluation.

Local gain exceeds uncertainty by 0.05637 nats, with the adjusted interval [0.04675, 0.06598]. The fixed disagreement subset contains 2,452 targets from 541 images; its difference is 0.04322 nats with adjusted interval [0.02023, 0.06593] (Figure 5). Both primary comparisons support a

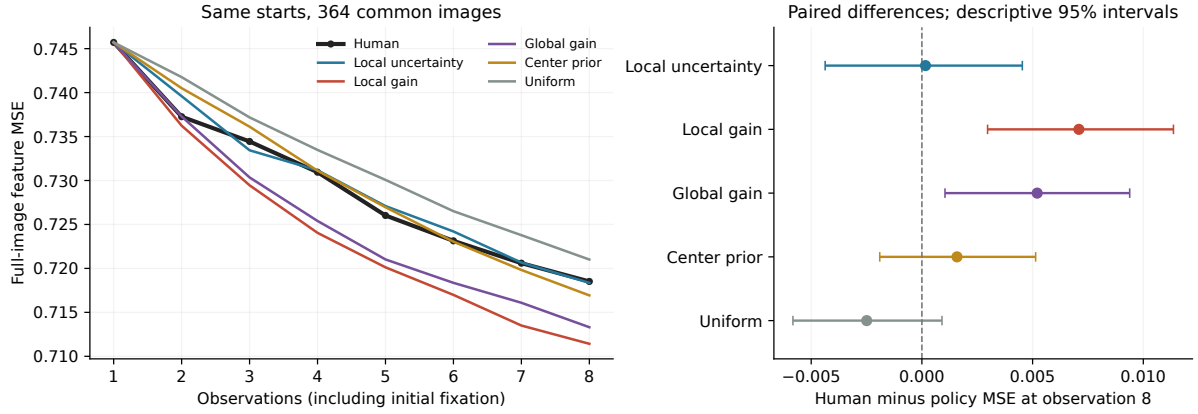


Figure 4: Human paths as inputs to the frozen scene predictor. Left: image-mean MSE on the fixed 364-image cohort, with one preselected human path per image and matched starts. The initial fixation counts as observation 1. Right: paired human-minus-policy terminal differences; negative favors human and positive favors the policy. Intervals are descriptive 95% image-bootstrap intervals without multiplicity correction. Curves are not a human next-choice likelihood evaluation, and movement and coverage differ (Table 5).

Table 6: Human next-cell prediction. Values are mean natural-log probabilities, first averaged within image; higher is better. Development scores are not independent final validation because upstream checkpoints used that split. DeepGaze has full-image access.

Model	Development (371 images)	Holdout (603 images)
DeepGaze III (full-image reference)	−3.91137	−3.85430
Controls + local gain	−4.15703	−4.13880
Controls + global gain	−4.19219	−4.17357
Controls + local uncertainty	−4.20753	−4.19517
Controls only	−4.21681	−4.20177
External center bias	−4.62596	−4.63631

local-gain predictive advantage. The effect is a difference in mean log score, not an increase in percentage accuracy or proof that each individual fixation follows gain.

In descriptive secondary comparisons, global gain exceeds uncertainty by 0.02160 nats, with a 95% interval of [0.01369, 0.02902]. Local gain exceeds global gain by 0.03477 nats, with interval [0.02855, 0.04135]. All fitted signal coefficients are positive. The full-image DeepGaze reference scores higher than all three value models (Table 6), showing that substantial predictable behavior remains outside these simple readouts. Its access and unresolved pretraining overlap prevent interpretation as an equal-access mechanism comparison.

4.7 Supported budgets and revisits

At budgets 1, 4, and 7, descriptive local-gain minus uncertainty differences are 0.07876, 0.04319, and 0.04671 nats. Each interval is positive (Figure 6). The values do not show a monotonic decrease with budget, and we have not performed a confirmatory budget-interaction test.

For next targets in previously unvisited cells, the difference is 0.06823 nats across 14,887 targets, with a 95% interval of [0.05872, 0.07752]. For returns to visited cells, it is −0.00698 across 2,950 targets from 598 images, with interval [−0.02211, 0.00938]. The latter does not clearly favor either signal. Thus, the development result favoring uncertainty on revisits does not clearly replicate in the holdout. These groups depend on the actual next target, so they are retrospective

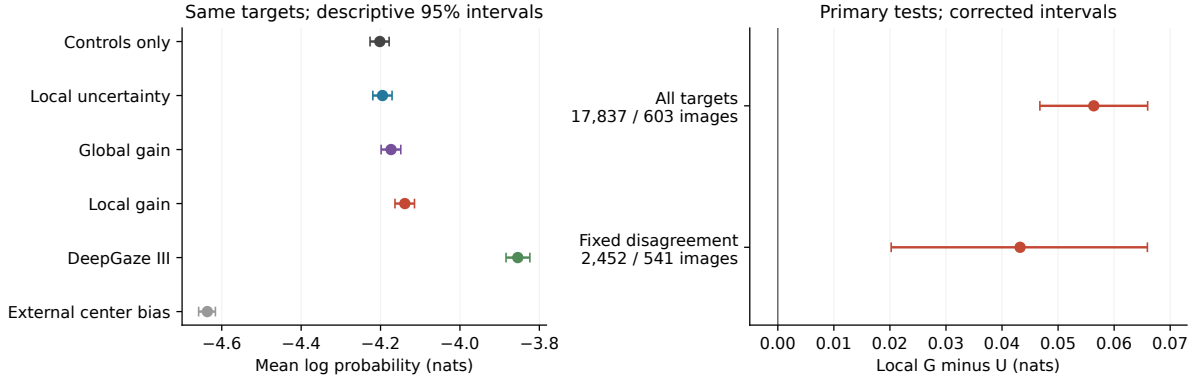


Figure 5: Frozen holdout results. Left: identical-target model scores with descriptive image-bootstrap 95% intervals; each U/G label denotes its controls-plus-signal model. Right: the two fixed primary comparisons, each with a 97.5% interval, adjusted as a 95% family. Positive values favor local gain. The disagreement subset uses only before-choice signals and approximate area/distance matching of the preferred candidates.

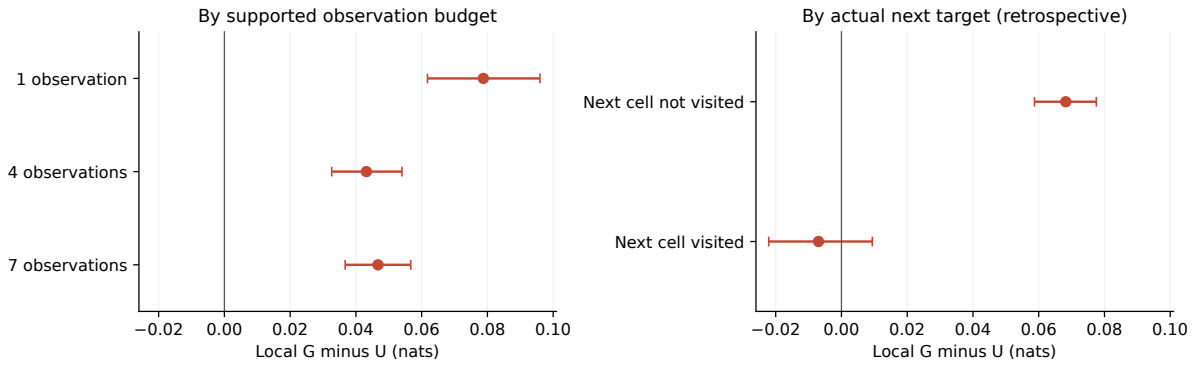


Figure 6: Descriptive holdout strata, with paired image-bootstrap 95% intervals. Local gain has positive differences at each supported budget. The actual revisit subset has an interval crossing zero. These comparisons are outside the two-test primary family; no additional subset search or fitting was performed on the holdout.

descriptions rather than a before-choice mechanism classifier.

4.8 Recovery supports finite-sample distinguishability

All 480 recovery fits pass the original convergence criterion. The uncertainty, local-gain, and global-gain generators are each correctly distinguished in all 30 replicates after pairwise multiplicity correction (Figure 7). In control-only simulations, no value model has supported incremental predictive value in any replicate. Yet selecting the highest mean score alone chooses a value model in 25 of 30 control-only replicates. Small numerical advantages among near-equivalent models should therefore not be treated as evidence for a value signal.

Finite replication limits this result. A 30/30 success rate has a Wilson 95% lower bound of about 88.65%; a 0/30 false-evidence rate has an upper bound of about 11.35%. Recovery concerns fitted signal strengths and recorded states. It does not prove universally low error rates, recovery of a full cognitive mechanism, or recovery of new scanpath dynamics.

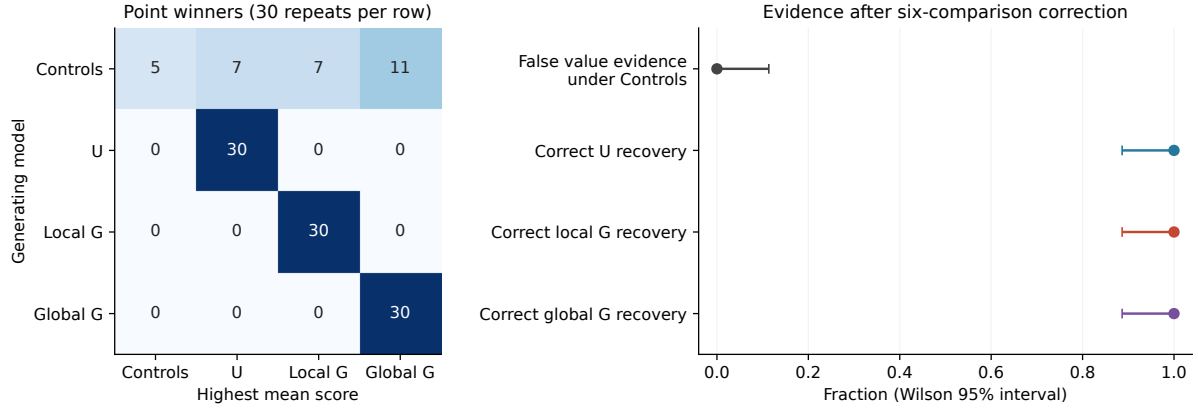


Figure 7: Model recovery on recorded states. Left: counts of the model with the highest mean development score, regardless of statistical separation. Right: evidence-based recovery for signal generators and false value evidence under the control-only generator. Error bars here are Wilson intervals across 30 replicates, not image-bootstrap intervals. The control-only model is not claimed to be uniquely recovered.

4.9 Later development diagnostics: candidate value and predictor capacity

Two subsequent diagnostics address why the closed-loop curves are close (Appendix F). They were conducted after the holdout results were known, using development images only. On identical center-policy histories, a hindsight oracle achieves a mean one-step global error reduction of 0.035314, compared with 0.003952 for the center prior and 0.006869 for learned global gain. Candidate locations therefore differ substantially in realized value under the current observer. The oracle sees information unavailable before a choice, so its advantage does not establish how much is learnable or identify the mean predictor as the main bottleneck.

A single deeper spatial-residual mean head improves full-image development MSE by 0.46% and new-region reveal benefit by 30.09%. It uses new patches more effectively, but misses the engineering adoption rule requiring more than 2% MSE improvement as well as improved reveal benefit. We retain the original observer and stop the architecture comparison. All closed-loop and human-choice results above continue to refer to that original observer; no new variance or gain model was fitted for the alternative.

5 Discussion

The completed study operationalizes one part of the broader resource-limited active-sensing program: valuing a new fixation by its expected effect on a specified visual prediction. U and G are candidate signals within that framework, not the entire definition of the project. The three branches allow functional usefulness, human predictive relevance, and descriptive benchmark performance to be reported separately.

The main finding is narrow but reproducible within this study: a local expected-gain signal predicts the next fixation better than local predictive uncertainty on new images, when both are read from the same fixed observer and enter comparable controlled choice models. The advantage remains in a subset where the preferred candidates differ but have similar new area and movement distance. This is evidence for the behavioral predictive relevance of an action-dependent signal beyond current uncertainty alone.

The distinction between local and global gain matters. Global gain is weaker both as a measured gain estimator and as a human predictor. This could reflect a difference in behavioral relevance, estimator quality, or both. The present results cannot conclude that people optimize local rather

than global error reduction. A fair mechanism comparison would require stronger evidence that both latent values are measured with comparable reliability.

The functional and behavioral results also should not be collapsed into one claim. Unrestricted gain policies reduce model error, which is the quantity they were trained to predict. Under a shared movement cap, terminal advantages become uncertain even though path-average error remains lower. Human predictive differences provide separate evidence, but they do not remove these functional qualifications.

Offline human-path replay provides a further connection between the two questions: real fixation locations do improve the same model’s scene prediction. Yet lower model MSE for gain policies does not imply poorer human sensing. Human paths reveal less area, move less, retain revisits, and were collected with full-image access. Functional replay and prediction of the human next action therefore remain distinct tests.

The later oracle and architecture diagnostics narrow the interpretation of these small functional differences. They rule out a simple claim that every available location has similar realized value on the tested states. They do not justify attributing the remaining gap entirely to a weak mean head: the one tested alternative improves patch use substantially but overall reconstruction only slightly. Other architectures remain untested, and the oracle includes hindsight information.

Several modeling assumptions limit interpretation. DINOv2 features define the target, so the result may depend on that representation. The variance head estimates residual dispersion around an imperfect mean. Gain magnitudes are biased and candidate rankings are only moderately accurate. Perfect memory makes exact repeated observations uninformative to this observer, although a human may benefit from a revisit because of forgetting, noise, or a changing goal. The original experiment did not impose our blur-and-patch access rule. We therefore do not claim to reconstruct each participant’s actual perceptual state.

Statistical control is not a causal intervention. The four choice models separately fit their control coefficients, and an omitted scene property could relate to both gain and fixation choice. Approximate matching of two preferred candidates does not equate whole choice sets. Gains used to fit the choice readout are produced on the gain model’s training images, rather than by cross-fitting; the frozen holdout evaluates the resulting complete pipeline but does not isolate this training dependency. Images are held out within the same participant sample, and pretrained encoders may have encountered related image content. No claim of pretraining-independent generalization is made.

6 Future work

The wider program can extend this static-image implementation to motion, fixation duration, limited or decaying memory, and stable observer differences. These are proposed extensions, not completed parts of the present study. A further distinct question is whether value predicts choices beyond a strong learned gaze reference: the current models compare against DeepGaze but do not include its log probability as a common covariate. Likewise, a future DeepGaze-driven closed loop would answer a new functional question and would require an explicit decision about its full-image access.

The completed holdout should remain a final evaluation of the current specification. A next study can test one explicit alternative explanation using a new evaluation arrangement: for example, changing the target representation or adding a controlled memory-decay model. Such changes should regenerate gain labels and refit the relevant models because gain is defined relative to the observer.

The one planned spatial-head comparison is complete and did not trigger replacement of the

observer. Further architecture searches and a new gain-training cycle are not part of the current study.

A stronger mechanism test would manipulate observation quality or cross-region predictability while keeping current uncertainty comparable. The Gaussian cases provide concrete designs for such dissociations, but their extension to human natural-image viewing remains untested. New participants and a gaze-contingent experiment could also test whether the assumed partial-access process captures the relevant evidence. These studies would address mechanisms more directly than further tuning for the same held-out prediction score.

7 Conclusion

Valuing a gaze decision requires an observation model, a prediction objective, a measure of benefit, and a resource constraint. Our current implementation combines a full-image behavioral reference, partial-view active sampling, and fitted conditional models of real human choices. Local gain has the lowest mean closed-loop error among tested policies and gives higher held-out human-choice likelihood than uncertainty, including a fixed disagreement subset. DeepGaze III remains the strongest evaluated human predictor and has not been tested as a closed-loop policy here. These results support a predictive-benefit approach to gaze value under the specified conditions, while leaving the broader questions of human goals, memory, individual differences, and causal mechanisms open.

Data and reproducibility

The analysis uses the public COCO-FreeView data and pretrained DINOv2 and DeepGaze III models cited above. No new human experiment was conducted. The source package contains the manuscript, references, figure-generation code, and provenance for all plotted results. Local experiment records preserve image splits, source and checkpoint hashes, configuration, and per-target scores. Raw images and gaze records are not redistributed with the manuscript. Authorship, affiliations, acknowledgments, and a public code-release destination are to be supplied by the research team before submission.

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A Implementation details

The mean network concatenates 384 partial-view feature channels, a reveal mask, a content-weight map, and two coordinate channels. Its body contains a 1×1 convolution from 388 to 128 channels, a 3×3 convolution with padding 1, and a 3×3 convolution with dilation 2 and padding 2. GELU follows each layer. A 1×1 output layer predicts 384 channels directly. There is no residual addition. The final 128-dimensional bottleneck remains; we do not transfer any numerical lower bound derived for an earlier model to this architecture.

The independent variance network receives partial features and detached mean predictions, plus the same four auxiliary channels, for 772 input channels. It uses the same 128-channel body structure and predicts 384 log-variance channels, clipped to $[-6, 4]$. Frozen mean output checks cover all 2,226 diagnostic development states.

The gain input has 2,309 dimensions: local and global means of partial features, predicted mean features, and log variances (six vectors of 384 values), plus candidate x, y , local revealed fraction, new content fraction, and budget divided by eight. A multilayer perceptron has widths $2309 \rightarrow 256 \rightarrow 64 \rightarrow 2$, with GELU hidden activations. Input and target normalization use training statistics. Training-label extraction samples human and random histories at budgets 1, 4, and 7, with eight candidates per training state and sixteen per development state. Candidate vectors contain only before-observation quantities. Before and after mean predictions use batch size one during final gain-label generation to avoid changing numerical behavior across the pair.

Table 7: Main fitting settings. Choices were fixed before holdout evaluation.

Component	Optimizer / learning rate	Batch	Stopping / selection
Mean	AdamW / 10^{-3} initially	8	Stop 70; select 60
Variance	AdamW / 3×10^{-4}	8	20 epochs; select 20
Gain	AdamW / 10^{-3}	512	10 epochs; select 2
Choice	Float64 L-BFGS	Image-weighted	200 iterations maximum

Mean training uses weight decay 10^{-4} , gradient norm clipping at 1, and learning-rate reduction by a factor of 0.5 after three plateau epochs, down to 10^{-5} . Early stopping uses a 10-epoch patience and relative improvement threshold 10^{-4} . The continuation after epoch 60 preserves optimizer and scheduler state and is capped at 20 extra epochs. Choice fitting allows at most 250 function evaluations. Tight internal tolerances retain the original acceptance requirement of gradient magnitude at most 10^{-6} ; failed preliminary fits were recomputed before final evaluation rather than accepted by relaxing that threshold.

B Choice controls and score aggregation

The 11 controls are $x, y, x^2, y^2, xy, d, d^2$, new visible area divided by total content area, revealed fraction within the candidate footprint, historical cell-visit count divided by t , and inverse lag since the last visit (zero if never visited). Here x, y are normalized content coordinates and d is distance from the last fixation in shorter-content-side units. Training standardization weights images equally, then states within image, then candidates within state. There is no intercept in the conditional softmax. Each signal model has its own control coefficients.

For image i with n_i eligible scored targets, the evaluation statistic is

$$\bar{\ell}_s = \frac{1}{N} \sum_{i=1}^N \frac{1}{n_i} \sum_{k=1}^{n_i} \log p_s(a_{ik} | h_{ik}). \quad (8)$$

Comparisons bootstrap the paired image contributions. For a subset, only images with at least one selected state enter that subset mean. Therefore, full-sample and subset effects need not average together according to their raw target counts. The score is for a content-conditioned grid cell, not a continuous fixation density or a probability of exactly reproducing the original pixel.

C Additional functional results

Table 8: Local/global gain minus uncertainty under the common movement cap. Negative differences indicate lower error. All intervals are descriptive paired image-bootstrap 95% intervals.

Comparison	Terminal MSE difference	Mean MSE over budgets 1–8
Local gain minus U	−0.003333 [−0.007245, 0.000723]	−0.003582 [−0.005845, −0.001173]
Global gain minus U	−0.001643 [−0.005808, 0.002308]	−0.002539 [−0.004905, −0.000224]

Under the cap, cumulative movement in shorter-side units is 1.043 for uncertainty, 1.156 for local gain, and 1.098 for global gain; corresponding coverage is 9.18%, 10.14%, and 9.49%. The cap derives from 481,422 valid original adjacent training transitions, without compressing histories across excluded border endpoints. An independent check verifies 33,761 selected transitions against the cap.

A retrospective geometry check on six fixed development images compares radii of 4%, 8%, and 12%. Average coverage after eight observations is 2.57%, 8.72%, and 16.94%, respectively. This is a geometry illustration, not an evaluation of retrained observers at other radii; it did not select or change the 8% model.

D A complete candidate-state illustration

Six development images were fixed by a hash ordering, with three histories per image, before generating full candidate diagnostics. This gives 18 states and 3,312 candidates. Figure 8 illustrates one previously inspected state. It shows why an expected map and a realized map are different objects: the latter uses information obtained after the hypothetical observation and is available only offline. Color scales are separate. A single state cannot establish calibration of a conditional expectation. The natural image is retained here for internal research illustration; the source package does not redistribute the underlying dataset.

E Verification and scope of the final comparison

Independent technical checks verify frozen model and protocol hashes; all 603 image identities; every target, raw history prefix, and candidate mapping; the before-choice disagreement rule; and both primary intervals recomputed from per-target records. The first scored DeepGaze state matches the official full forward pass with maximum log-probability difference zero. The complete holdout job takes approximately 97 minutes; subsequent numerical audit and figure jobs complete successfully. These are implementation checks, not additional evidence for a cognitive mechanism.

The main analysis plan is fixed before holdout scoring but after development exploration. Budget and actual-revisit strata are descriptive. We preserve the null revisit comparison and the uncertain constrained terminal comparison. No observer, gain model, choice coefficient, or subgroup threshold is adjusted after the holdout results are observed.

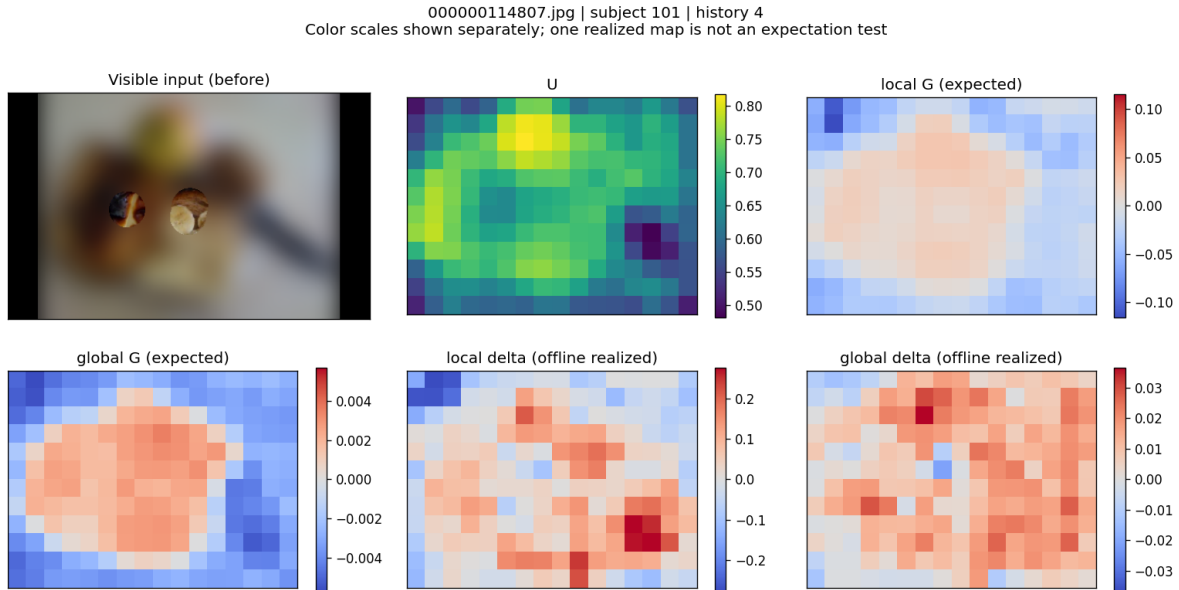


Figure 8: Example from development image 114807, participant 101, after four observations. Top left: the modeled current RGB view. Remaining panels show uncertainty, learned expected local/global gains, and offline realized gains over all candidate cells. The example illustrates definitions, not a selected statistical success. A learned expected gain need not match one realized outcome.

E.1 Additional integrity audit and a numerical reproducibility limitation

A subsequent read-only audit verifies archived source-data hashes and all 4,317 image hashes. Independent NumPy calculations reproduce the four choice-model training objectives on all 99,004 training targets, with maximum objective discrepancy below 5.4×10^{-15} and gradients below the original 10^{-6} criterion. Recomputing all 17,837 holdout target probabilities from cached candidate features and fitted coefficients differs from stored values by at most 1.8×10^{-15} . Both complete development closed-loop MSE curves are reconstructed exactly from trajectory records. No outcome-specific score adjustment or fabricated human record was identified in the inspected pipeline. These checks support provenance and internal reproducibility, not an absolute certification of data collection or every historical analysis.

A stronger, fresh image-to-network check does *not* meet its initial numerical tolerance. On two fixed holdout images and three states per image, the maximum absolute candidate differences are approximately 0.000973 for U, 0.000244 for local G, and 0.000070 for global G; geometric and history controls match exactly. The maximum target log-probability difference is 0.003234 nats for the value models and 0.000444 nats for DeepGaze. Four sampled training gain labels differ by at most 0.000775. The original evaluation used an A100 MIG allocation; the short audit used A100 MPS under cluster rules. The weights and source are fixed, but matching-execution-configuration validation has not established the cause of the discrepancy. GPU configuration and numerical kernels are a plausible explanation, not a proven diagnosis. The original tolerance failure is retained.

Thus the saved-feature statistical results are independently reproducible, while exact end-to-end numerical reproduction has not been demonstrated. The six-state sample is insufficient to guarantee unchanged whole-dataset scores or near-tied argmax choices. We did not retrain every model or rerun every calibration candidate, and we do not claim hardware-independent bitwise reproduction. This limitation also motivates recording the execution configuration in future evaluations.

E.2 Explicit modeling decisions are not empirical observations

Patch size, blur, grid resolution, network widths, learning rates, fixed seeds, and regularization are chosen parameters. Mean architecture changes and checkpoint selection used development results. The disagreement thresholds and primary local-gain contrast were fixed after development exploration, before holdout scoring. Variance adequacy was a documented engineering judgment without a preregistered numerical pass threshold. These choices are disclosed rather than presented as assumption-free or confirmatory findings.

The Gaussian covariance/noise cases are constructed examples; model-recovery choices are simulated from fitted models. Neither is a source of additional human observations. Exact zero gain for a no-new-pixel repeat is an explicit consequence imposed by the deterministic perfect-memory contract, not an independently learned finding. Reporting tables transcribe verified results; plotted empirical curves read saved results rather than embedding a desired policy ranking. The separate audit report records the scope, failure, and diagnostic artifacts.

F Later development diagnostics

These analyses were motivated after inspecting the completed study, including its holdout results. They use the original 371 development images and no holdout examples for computation or fitting. They are exploratory diagnostics, not new independent confirmation of the human-choice findings. Their addition does not resolve the numerical reproducibility limitation in Appendix E.

F.1 One-step hindsight oracle on identical histories

Run 22547 reuses center-prior seed-0 histories from the original unrestricted closed loop at budgets 1, 4, and 7: 1,113 states across 371 images. At each state, it evaluates every candidate that reveals new pixels using the original frozen observer, totaling 200,997 candidate reveals. The global oracle selects the candidate with the largest realized full-image error reduction, using the full target and the result after revealing each patch. U and learned G select using only their original before-choice signals. Uniform and center-prior values are exact probability-weighted expectations over the eligible candidates. The same candidate records also support a comparison under the existing training-derived movement cap.

Table 9: Mean one-step full-image MSE reduction on identical development histories. Higher is better. These are counterfactual next choices from fixed center-policy states, not eight-step closed-loop endpoints. The hindsight oracle is not an implementable before-choice policy.

Selection rule	Unrestricted	Common movement cap
Uniform expectation	0.003223	0.003429
Center-prior expectation	0.003952	0.003524
Uncertainty U	0.004494	0.003810
Learned local G	0.006528	0.005278
Learned global G	0.006869	0.005238
Hindsight global oracle	0.035314	0.023430

The unrestricted oracle-minus-center difference is 0.031362, with descriptive 95% paired-image bootstrap interval [0.029238, 0.033715]; the corresponding capped difference is 0.019906, [0.018596, 0.021257]. There are no stalled states. These results establish substantial realized candidate variation on the tested histories. They do not establish that the entire oracle advantage can be predicted from a partial view. In particular, this comparison cannot by itself distinguish errors in the mean predictor, errors in the gain estimator, and information that is unavailable until the new patch is seen. The differing local/global G order here and in the eight-step experiment also concerns different histories and evaluation horizons.

F.2 A single spatial-residual mean comparison

Run 22548 keeps the frozen DINO encoder, normalization, image split, and MSE objective. It replaces the shallow mean head with a $388 \rightarrow 128$ projection, four internal residual blocks with two 3×3 convolutions each and dilations 1/2/4/1, and a direct $128 \rightarrow 384$ output. Internal skip connections do not add the partial feature map to the final prediction. The 128-channel output bottleneck remains.

Training starts from scratch with AdamW, initial learning rate 0.001, batch size 8, the original learning-rate decay and early-stopping settings, and at most 80 epochs. It stops at epoch 62 and selects epoch 60. The existing baseline is reused. Comparison records match all 2,226 development states and 17,808 reveal candidates. New-region benefit measures before-minus-after

error specifically on newly revealed pixels; a mask-only control changes the reveal mask without supplying the patch content.

Table 10: The single alternative mean head. Both metrics improve, but only patch benefit meets the minimum relative improvement specified before this run’s results. The development set also selects checkpoints, so intervals are descriptive.

Metric	Original mean	Spatial mean	Relative improvement
Full-image MSE (lower better)	0.732221	0.728882	0.46%
New-region benefit (higher better)	0.024748	0.032194	30.09%

Paired MSE improvement is 0.003338, [0.000131, 0.006508], and paired new-region benefit increases by 0.007446, [0.005555, 0.009359]. Correct patch content outperforms the mask-only control by 0.033177 on the new region, [0.030392, 0.035997]. All intervals are descriptive 95% image-bootstrap intervals.

Before this run’s results, the engineering rule required more than 2% relative MSE improvement and at least 10% new-region benefit improvement, positive paired lower bounds, and a positive patch-content control. These are explicit adoption thresholds, not universal scientific criteria or a preregistration of the whole study. The MSE threshold is missed. We therefore retain mean run 22455, variance run 22459, and G run 22477; the new checkpoint remains a comparison artifact. This is a small overall improvement with a meaningful gain in patch use, not evidence that the alternative is useless or that stronger predictors cannot help.

The source package includes the oracle summary, architecture decision, and fixed architecture protocol under `diagnostic_evidence/`, with hashes and original run identifiers. This writing update checks those saved records and source bindings; it does not rerun the two experiments or extend the earlier integrity audit to all their numerical outputs.